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Introduction

Preterm infants are an extremely vulnerable population and an increased exposure to light and noise negatively influences their stress levels and development. Increased light and sound levels also put staff and parents at risk of psychological stress. High light and sound levels in NICUs (Neonatal Intensive Care Units) are frequent and efficacious interventions aiming at reducing light and sound levels rely on structural, cognitive and educational components.

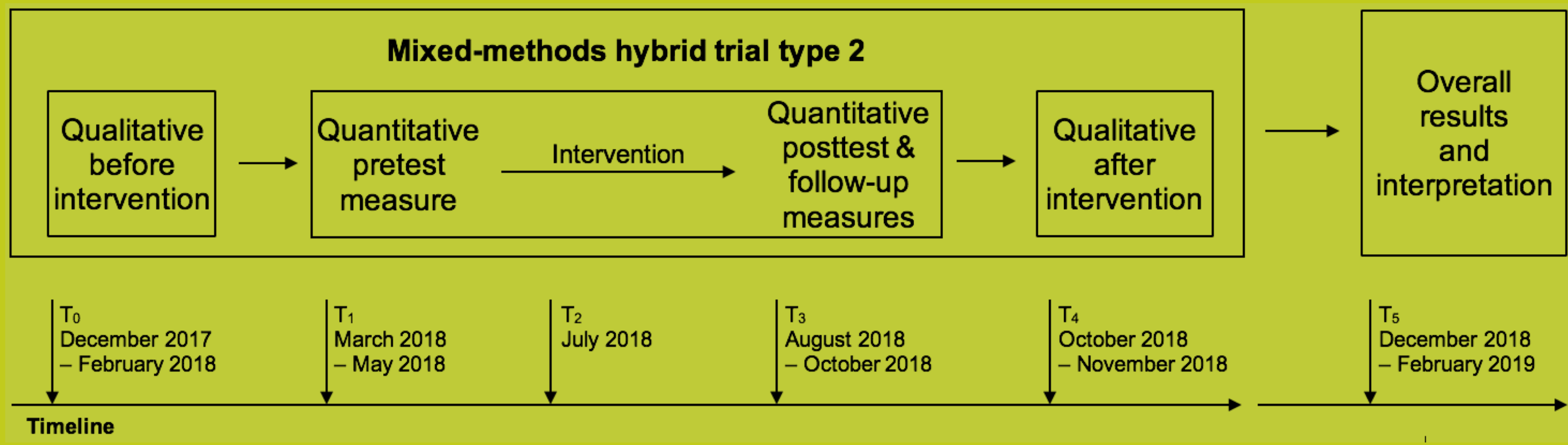
Aims

1. Reduce the light and sound levels on the NICU through development and implementation of a light and sound reducing intervention.
2. Reduce the light and sound induced stress-levels of premature children on the NICU.

Methods

A Hybrid Trial Type 2 study with a pre-test – post-test design allowed a combined analysis of clinical effectiveness and implementation of the intervention. The main steps of the study process were: (1) Adapting and planning intervention with focus group; (2) Baseline assessment through questionnaires, on-ward measurements and structured observation rounds; (3) Evaluation of outcomes after the intervention; (4) Assessment of implementation with focus group (Figure 1).

Figure 1. Study design



Action research was used to develop the intervention in a participatory effort and to increase acceptability. Focus groups were conducted with 4 expert nurses and 1 expert physician, all working on the NICU.

Data was collected from multiple sources at baseline and at 1-month after the intervention over a 4-week period. Light and sound levels were measured in the room, at patients’ bedside and in incubators using approved devices. Patient information (age, gestational age, main diagnosis), clinical data (oxygen-needs, heart and respiratory rate, frequency of desaturation, bradycardia and apnoea) and contextual data (number of patients, machines and equipment, grade-mix) were collected from patient files and on-site by trained staff. A questionnaire was distributed to all professionals working on the NICU (nurses, nurse assistants and physicians) to assess knowledge, common practices, readiness to change and satisfaction. The questionnaire was anonymous and returning it implied consent.

Results

The intervention developed had structural, cognitive and behavioural components (Table 1). Most importantly, a 15-minute workshop including an educational film sequence was developed.

Table 1. Detailed components of intervention

	Components (active ingredients)	Mode of delivery
Structural	Reduction of light and sound levels of equipment	- Technician
	Elaboration of a standard of procedure (SOP)	- Introduce new staff members to SOP systematically
Document-ation (cognitive)	Presentation to staff	- Routine presentation at nursing and medical team-meetings or colloquiums
		- All staff over a 4-week period
Educational (behavioural)	Educational prog. (30-min workshop):	- Routine part of team-meeting
	- Present light and sound levels on unit	- Training new staff members (individually or in session)
	- Value/emphasise efforts done	- Accessible through intranet any time
	- Possibilities for improvement	- Applied in patient room and visible at all times
Feed-back (cognitive)	Short film and slide presentation	- Scale in ward entrance area, accessible to parents and visible to parents and staff
	Noise-sensor light alarm	- Present summary at team-meeting, colloquium, e-mail
	Patient family driven scale indicating subjective appreciation of light and sound levels	
	Summarise results and analysis of measurements (light and sound levels) and focus group observations	

Table 2. Patient characteristics

Variables		Pretest	Posttest	Differences ¹
		Values/ scores	Mean ± SD	
			Mean ± SD	
			N (%)	
Socio-demographic				
Patients ²	<i>N/day</i>	4.2 ± 0.8	3.8 ± 0.7	<i>t</i> = 1.8, <i>p</i> =.09
Gender ³ (female)	<i>N/day</i>	0.8 (17.3)	2.4 (59.4)	<i>t</i> = -6.7, <i>p</i> <.05
Gestational age	<i>wk</i>	27.8 ± 0.9	27.8 ± 1.6	<i>t</i> = -0.1, <i>p</i> =.91
Birth age	<i>days</i>	19.8 ± 5.6	27.2 ± 14.0	<i>t</i> = -2.7, <i>p</i> <.05
Weight	<i>g</i>	1207.8 ± 278.4	1412.8 ± 431.1	<i>t</i> = -2.1, <i>p</i> <.05
Clinical				
O ₂ needs	<i>FiO₂</i>	0.28 ± 0.04	0.28 ± 0.08	<i>t</i> = 0.37, <i>p</i> =.71
SpO ₂ falls	<i>N/day</i>	45.9 ± 17.5	25.7 ± 18.3	<i>t</i> = 4.2, <i>p</i> <.05
HR falls	<i>N/day</i>	4.3 ± 3.1	7.4 ± 3.3	<i>t</i> = -3.6, <i>p</i> <.05
Apnoea	<i>N/day</i>	0.6 ± 0.9	1.0 ± 1.6	<i>t</i> = -1.2, <i>p</i> =.23

Note. ¹Significance of differences measured pretest and posttest. ²Max N = 6. ³Difference in proportions.

Table 3. Light and sound levels

Area	Pretest			Posttest		
	Shift 1	Shift 2	Shift 3	Shift 1	Shift 2	Shift 3
Light (lux, median, Q2-Q4)						
Incubator	24.6 (8.7-43.8)	0.0 (0.0-2.3)	0.7 (0.0-1.3)	4.2 (3.1-7.6)	4.7 (0.0-8.0)	0.0 (0.0-0.2)
Bedside	39.4 (28.1-47.7)	29.6 (6.5-77.8)	1.9 (1.1-4.5)	41.0 (12.0-85.0)	6.0 (0.0-46.0)	0.0 (0.0-1.0)
Room	111.0 (54.0-174.0)	5.0 (0.0-43.0)	0.0 (0.0-1.9)	39.0 (12.0-85.0)	3.0 (0.0-34.0)	0.0 (0.0-0.0)
Sound (dB, mean ± SD)						
Incubator	46.2 ±4.9	44.4 ±3.2	44.5 ±3.8	45.8 ±3.4	45.7 ±3.3	44.6 ±3.7
Bedside	51.6 ±5.7	50.7 ±5.5	49.1 ±5.5	51.4 ±5.9	50.3 ±5.3	48.3 ±4.9
Room	52.3 ±6.0	50.9 ±5.8	49.3 ±5.8	52.5 ±6.0	53.2 ±6.3	49.9 ±5.9

A reduction of light and sound levels over most shifts was observed after the intervention and was significant for light levels on day-shifts (*t* = 2.7, *p* <.05; Table 3). After controlling for confounding factors, the intervention seemed significantly associated with lower sound levels (*β* = -2.6, *p* <.001), the association with light levels was not significant.

Discussion

Findings support the clinical effectiveness (fewer desaturations) of an intervention based on structural, cognitive and behavioural components. The participatory design contributed to a high involvement of the staff and a strong implementation of interventions to reduce sound and especially light levels on a NICU.

A major strength of this study was to include physicians in the development and parents in the implementation and evaluation of the intervention, to endorse the change of culture.

The model performed better and explained up to 80% of sound levels’ variability which may be related to a more comprehensive scope of literature relating to sound levels.

A multicentre randomised design would have allowed for stronger assertions towards the clinical effectiveness of the intervention.

Contact (for further references)

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